

Absolute Maximum Ratings (Note 1)

If military/aerospace specified devices are required, contact the National Semiconductor Sales Office/Distributors for availability and specifications.

	LF155/6	LF256/7/LF356B	LF355/6/7
Supply voltage	±22V	±22V	±18V
Differential input voltage	±40V	±40V	±30V
Input voltage range (Note 2)	±20V	±20V	±16V
Output short circuit duration	Continuous	Continuous	Continuous
T_{JMAX}			
H-package	150°C	115°C	115°C
N-package	—	100°C	100°C
M-package	—	100°C	100°C
Power dissipation at $T_A = 25^\circ\text{C}$ (Notes 1, 8)			
H-package (still air)	560 mW	400 mW	400 mW
H-package (400 LF/min air flow)	1200 mW	1000 mW	1000 mW
N-package	—	670 mW	670 mW
M-package	—	380 mW	380 mW
Thermal resistance (typical) θ_{JA}			
H-package (still air)	160°C/W	160°C/W	160°C/W
H-package (400 LF/min air flow)	65°C/W	65°C/W	65°C/W
N-package	—	130°C/W	130°C/W
M-package	—	195°C/W	195°C/W
(Typical) θ_{JC}			
H-package	23°C/W	23°C/W	23°C/W
Storage temperature range	-65°C to +150°C	-65°C to +150°C	-65°C to +150°C
Soldering information (lead temp.)			
Metal can package			
Soldering (10 sec.)	300°C	300°C	300°C
Dual-in-line package			
Soldering (10 sec.)	260°C	260°C	260°C
Small outline package			
Vapour phase (60 sec.)	—	215°C	215°C
Infrared (15 sec.)	—	220°C	220°C
See AN-450 'Surface Mounting Methods and Their Effect on Product Reliability' for other methods of soldering surface mount devices.			
ESD tolerance (100 pF discharged through 1.5k Ω)	1000V	1000V	1000V

Absolute maximum ratings tell the designer the conditions that will damage the device. Remember these are not maximum operating limits

Fig. 4: Absolute maximum ratings for op-amp type LF155/256/356

As you can see from the figure, it is available in three packages, namely, the metal can package, dual-in-line package and ceramic flatpak package.

Schematic diagram. Detailed schematic of the internal structure of the device is also given in the datasheet. Fig. 3 shows the internal schematic of operational amplifier LM741. It helps the designer to correlate the electrical features of the device with its internal

architecture. This sometimes helps the designer to explain the device behaviour that is otherwise not described in the datasheet.

Absolute maximum ratings. Absolute maximum ratings (Fig. 4) contain information on voltages, currents, powers and temperature range that should never be exceeded. As is evident from the figure, the maximum supply voltage for operational amplifier LF356 is ±18V.

KEPTRON

The House of Power Technology

AN ISO 9001:2000 CERTIFIED COMPANY

DOUBLE CONVERSION

PURE SINE WAVE

Online UPS



1 KVA to 100 KVA

Latest State-of-the Art PWM Controlled IGBT Based design

High efficiency and reliability

Low cost

OTHER PRODUCTS

LINE-INTERACTIVE UPS
(DC 12V to 48V) : 500VA to 1 KVA

INVERTER (SINE-WAVE) :
300 VA to 50 KVA

HOME UPS : 300 VA to 1 KVA

SERVO STABILIZERS :
2 KVA to 500 KVA

Battery Charger (SMPS)

KEPTEK POWER CONTROLS PVT. LTD.

110-A, Palpanganj Village, Near Community Centre, Delhi-110091 (India)

Tel.: 011-22751719, 22755542, 65292160

Telefax: 91-11-22793019

Mob.: 9810046719, 9958923180

E-mail: keptron@vsnl.net, keplekimd4.vsnl.net.in

Dealer's Enquiry Solicited